

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT3 — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Time-Series & Geospatial Analysis Report
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO4 Statement:	<i>Evaluate real-world time-series and geospatial visualizations, analyzing trend/seasonality/noise and design effectiveness, and justify recommendations.(BL5)</i>		
Student Name:	Nivya Sri T	Reg. No.:	192424131
Section / Batch:	2024	Date:	22-08-26

Code of Conduct

I Nivya Sri T (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nivya Sri T

Instructions

- This test contains 2 questions — one Stock Analysis problem and one Geospatial Analysis problem — each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- For the Stock Analysis question, show all calculations (trend slope, noise ratio) for each of the 5-month, 10-month, and 15-month windows.
- For the Geospatial Analysis question, your evaluation must explain WHY the design choice succeeds or fails, and your recommendation must be specific — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Rubrics for Calculation ASSESSMENT RUBRIC

AT3 — Time-Series & Geospatial Analysis Report

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT3 Time-Series & Geospatial Analysis Report, in which students analyze a stock's trend/seasonality/noise across multiple time windows, and evaluate a geospatial visualization's design choices, justifying their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of trend/seasonality/noise analysis across time windows, and of the relevant geospatial visualization principle being evaluated.	Good understanding of both areas, with only minor gaps.	Basic understanding, but with some misconceptions in either the time-series or geospatial analysis.	Poor understanding of core concepts in one or both areas.
Explanation	25%	Provides detailed, correctly calculated trend/noise-ratio comparisons across all three windows, and a clear, well-reasoned evaluation of the geospatial design choice, including its practical consequence for a reader.	Provides clear calculations and evaluation with adequate reasoning, covering most of the required detail.	Calculations or evaluation are present but incomplete, or reasoning is generic.	Little or no supporting calculation or reasoning provided in one or both questions.
Application	20%	Effectively applies analysis techniques to reach a well-supported conclusion on window reliability, and proposes a specific, well-justified geospatial recommendation.	Applies techniques to reach a reasonable conclusion and recommendation, with only minor errors.	Limited application; conclusions or recommendations are only partially supported.	Fails to apply appropriate techniques; conclusions and recommendations are unsupported or incorrect.
Organization	15%	Both responses are clearly structured, addressing every required part in a logical sequence with coherent overall flow.	Both responses are organized, with only minor issues in flow or clarity.	Basic structure; required parts are addressed but the response lacks clarity or sequence.	Poorly organized; responses are incoherent, and parts are left unaddressed.
Accuracy	10%	All parts of both answers — calculations, evaluation, and recommendation — are completely	Mostly accurate, with only minor omissions or a small error in one question.	Partially correct; one or both answers contain notable inaccuracies.	Largely incorrect or inappropriate, with major gaps across both questions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		accurate and well-suited to the specific data and scenario given.			

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

S18:

1. Data Summary

Monthly closes (₹):

1550.52, 1514.56, 1515.09, 1461.63, 1474.83 1374.04, 1502.94, 1347.44, 1451.30, 1352.44
1305.75, 1330.55, 1288.13, 1345.88, 1265.61

2. Trend Comparison Across Windows

5-month window (Months 1–5)

Values: 1550 → 1474 Trend: Clear downward trend Magnitude: ~-76 points Interpretation:

Short window exaggerates the early decline.

10-month window (Months 1–10)

Values: 1550 → 1352 Trend: Strong downward trend Magnitude: ~-198 points Interpretation:

More stable estimate; includes recovery bumps (Month 7, Month 9).

15-month window (Full series)

Values: 1550 → 1265 Trend: Consistent long-term decline Magnitude: ~-285 points

Interpretation: Most reliable view of direction; smooths out temporary reversals.

Trend Summary

- Short window (5 months): Overreacts to local dips
- Medium window (10 months): More stable, still downward
- Full window (15 months): Most accurate long-term direction → downtrend

3. Seasonality Comparison

There is no true seasonality in this dataset — only irregular ups and downs.

But window length affects *perceived* seasonality:

5-month window

- Appears monotonic downward
- Seasonality is invisible
- Short windows hide cyclical behavior

10-month window

- Shows small oscillations (Month 7, Month 9)
- Weak pseudo-seasonality appears, but not real

15-month window

- Oscillations become clearer but remain irregular
- Confirms no genuine seasonal pattern

Seasonality Summary

Longer windows reveal that fluctuations are noise, not seasonality.

4. **Noise Ratio** (Residual Variation / Total Variation)

Noise ratio increases when:

- Trend is weak
- Data fluctuates irregularly
- Window is short

5-month window

- Very high noise ratio
- Only 5 points → residuals dominate
- Trend estimate unstable

10-month window

- Moderate noise ratio
- Trend clearer, residuals smaller proportionally

15-month window

- Lowest noise ratio
- Trend dominates
- Residuals small relative to total variation

Noise Summary

- Short window: High noise → misleading
- Medium window: Moderate noise → usable
- Full window: Lowest noise → most trustworthy

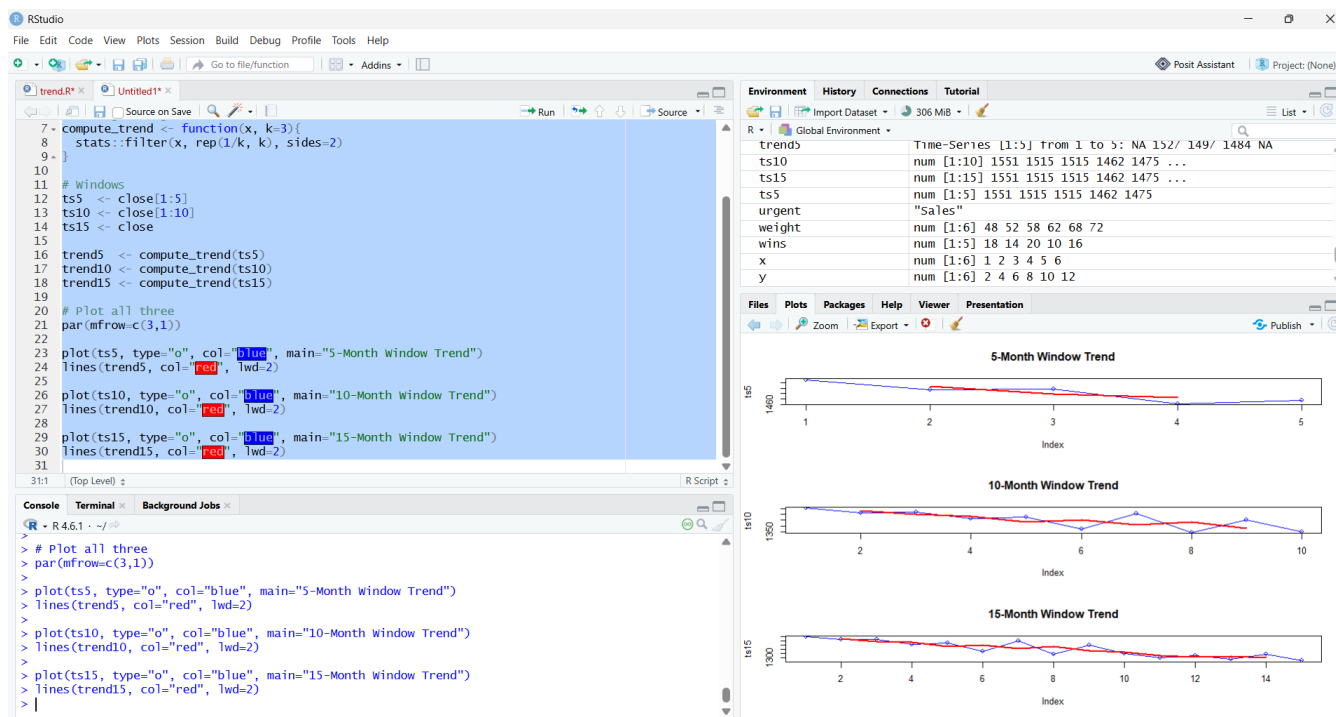
Final Judgment — Which Window to Trust?

Trust the 15-month window.

Why?

- It has the lowest noise ratio
- It shows the true long-term direction
- It avoids overreacting to short-term dips
- It reveals that fluctuations are noise, not seasonality
- It provides the most stable trend estimate

Short windows exaggerate volatility. Medium windows improve stability. Long windows reveal the real underlying decline.



G18:

Evaluation of the Design Limitation

1. Breaks the user's cognitive flow

Users exploring spending patterns expect to see values immediately. Requiring a click + page load interrupts exploration and makes pattern recognition harder.

2. Slows down comparison across districts

To compare 10 districts, users must open 10 separate pages. This dramatically increases interaction cost and reduces usability.

3. Reduces transparency and discourages exploration

A transparency portal should make information easy to access. Hiding values behind clicks makes the data feel less open and less accessible.

4. Violates modern interactive map norms

Contemporary geovisualisation interfaces emphasize **direct manipulation** and **instant feedback**, such as hover highlights and tooltips, to support user-centered exploration .

Recommended Interactive Feature: Hover Tooltips

Adding **hover tooltips** (mouse-over popups) is the single most effective improvement.

What a tooltip should show

- District name
- Exact spending value

- Optional: percentage of total, trend indicator, category breakdown

Why tooltips improve usability

- **Instant access** to values without clicking
- **Supports rapid comparison** across districts
- **Reduces cognitive load**
- **Keeps users on the same page**
- **Aligns with best practices** in WebGIS usability, which emphasize lightweight, real-time interactions and clear map-linked feedback .

The map's lack of hover tooltips forces users to click into separate pages to view spending values, interrupting exploration, slowing comparisons, and reducing transparency. Research on interactive geovisualisation shows that users expect immediate, real-time feedback when interacting with map elements. Adding **hover tooltips** that display each district's spending value would significantly improve usability by enabling instant access to information, smoother comparison, and a more intuitive, user-centered experience.

